2018 E.C BAHIR DAR CITY ADMINISTRATION MATHEMATICS MODEL EXAMINATION FOR GRADE 12

Time Allowed: 3 hours.

Instruction: There are 60 multiple choice questions, all have four alternatives. Choose the correct answer from the given alternatives and write your answers on the separate answer sheet.

1. Suppose A and B are two sets, where $AnB' = \{5, 2, 0\}$ and $A'nB = \{4, 1\}$. Which of the following is equal to AnB?

   A. $\emptyset$ C. $\{0, 1, 2, 4, 5\}$

   B. $\{0, 2, 4\}$ D. $\{1, 4\}$

2. What is the interval notation of $\{x \in \mathbb{R}: 9 > -x \ and \ x \neq 5\}$?

   A. $(-\infty, 9)$ C. $(9, \infty)$

   B. $(-9, 5)\ U\ (5, \infty)$ D. $(-9, \infty)$

3. Which of the following describes the property of composite numbers?

   A. Each composite number has more than one prime factor.

   B. Each composite number has at least two odd factors.

   C. All composite numbers are even numbers except 9 and 15.

   D. Every composite number has three or more different factors.

4. A farmer sold 87.86 kg of corn when measured correct to two decimal places. Which of the following is a lower bound of this weight in kg?

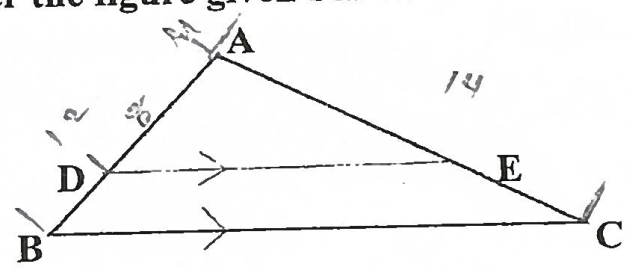
   A. 87.86 B. 87.855 C. 87.85 D. 87.859

5. Which one of the following is not true?

   A. All isosceles triangles are similar C. Every congruent polygons are similar

   B. All equilateral triangles are similar D. All regular pentagons are similar

6. Consider the figure given below.



Let $AB = 12$

$AD = 8$

$AC = 14$

$EC = x$ and $\overline{DE} \parallel \overline{BC}$

Then what is the value of x?

A. 5 B. $\frac{8}{3}$ C. $\frac{4}{3}$ D. $\frac{14}{3}$

7. If $kx^3 + 2x^2 - 3kx + 2$ is divisible by $x + 1$, then the value of k is:

   A. -1 B. 2 C. -2 D. 1

8. Find the value of x if $\log_2\left(1+\frac{1}{x}\right)=3$.

A. $\frac{1}{7}$ B. $\frac{-1}{7}$ C. 7 D. -7

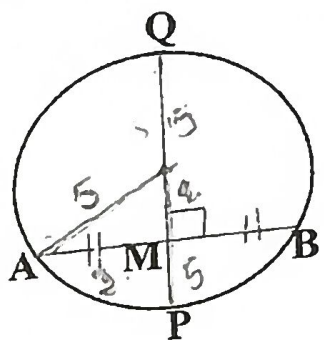
9. Which one of the following inequalities is true?

A. $tan(370^o)<0$ C. $sin(-100^o)<0$

B. $cos(200^o)>0$ D. $cos(-180^o)>0$

10. In the figure given below, the circle has radius $5cm$ and the length of chord $\overline{AB}$ is $6cm$. Moreover $\overline{PQ}$ is the perpendicular bisector of $\overline{AB}$.

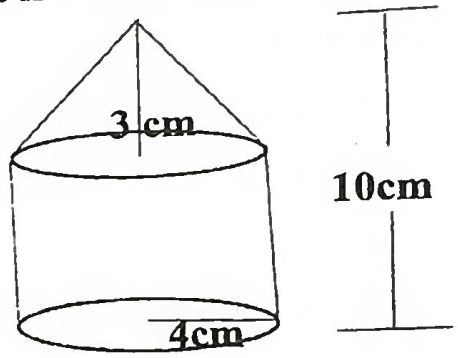
Which of the following equals the distance from Q to $\overline{AB}$?



A. $9cm$ C. $6\sqrt{2}cm$

B. $8cm$ D. $\sqrt{11}cm$

11. A solid figure is made of a circular cylinder of radius $4cm$ at the bottom and a right circular cone of altitude $3cm$ as shown in the figure below. If the overall height is $10cm$, then what is the volume of the solid in cm^3?



A. 64π C. 128π

B. 256π D. 120π

12. A point divides internally the line segment joining the points $A(-7,4)$ and $B(8,9)$ in the ratio 3: 2. What are the coordinates of a point?

A. $(2,7)$ B. $(-2,3)$ C. $(2,5)$ D. $(7,2)$

13. Let $R=\{(x,y): x-y \text{ is an integral multiple of } 2\}$. Which of the following belong to R?

A. $(-1,2)$ B. $(4,7)$ C. $(5,2)$ D. $(-1,-5)$

14. If $h=\{(1,2),(2,3),(3,5)\}$ and $g(x)=\frac{3}{2}x^2+\frac{1}{2}x$, then what is the value of $\left(\frac{g}{h}\right)(3)$?

A. Cannot be determined B. 5 C. 15 D. 3

15. The solution of the rational inequality given by $\frac{2x}{x-4}\leq 1$ is:

A. $-4\leq x\leq 4$ C. $\mathbb{R}$

B. $-4\leq x<4$ D. $-4x<4$

16. Decomposing $\frac{1}{x^2+3x}$ into its partial fraction gives:

A. $\frac{1}{3x}+\frac{1}{3x+9}$ C. $\frac{1}{3x+9}+2$

B. $\frac{1}{3x}-\frac{1}{3x+9}$ D. $\frac{-1}{3x}+\frac{1}{3x+9}$

17. If A and B are two matrices such that $A+B=\begin{pmatrix}7 & 0\\ 2 & 5\end{pmatrix}$ and $A-B=\begin{pmatrix}3 & 0\\ 0 & 3\end{pmatrix}$, then matrix A equals:

A. $\begin{pmatrix}10 & 0\\ 2 & 8\end{pmatrix}$

B. $\begin{pmatrix}2 & 0\\ 1 & 4\end{pmatrix}$

C. $\begin{pmatrix}5 & 0\\ 1 & 4\end{pmatrix}$

D. $\begin{pmatrix}5 & 4\\ 0 & 1\end{pmatrix}$

18. If $\begin{pmatrix}2 & -1\\ a & 4\end{pmatrix}\begin{pmatrix}1 & b\\ -1 & 3\end{pmatrix}=\begin{pmatrix}3 & 1\\ -1 & 10\end{pmatrix}$, then what are the values of a and b, respectively?

A. 2 & 3 B. 2 & -2 C. -1 & 2 D. 3 & 2

19. Let $A=\begin{pmatrix}1 & x\\ 2 & x+1\end{pmatrix}$. If $det(A)=0$, then what is the value of x?

A. 0 B. -1 C. 1 D. 2

20. Let A be a square matrix of order 3. If $det(A)=4$, then what is the value of $det(2A)$?

A. 8 B. 16 C. 24 D. 32

21. If $\overline{V}=\overrightarrow{AB}+3\overrightarrow{BA}$ where A and B are distinct points in the coordinate plane, then which one of the following is equal to $3\overline{V}$?

A. $-6\overrightarrow{AB}$ B. $6\overrightarrow{AB}$ C. $12\overrightarrow{AB}$ D. $-12\overrightarrow{AB}$

22. If $\vec{A}$ and $\vec{B}$ are parallel vectors with opposite directions and $|\vec{B}|=|2\vec{A}|$, then $\vec{B}-\vec{A}$ is:

A. $\vec{A}$ B. $-\vec{A}$ C. $3\vec{A}$ D. $-3\vec{A}$

23. A translation T takes the point $(1,0)$ in to $(4,2)$. Which of the following is the equation of the image line l: $3x+2y-5=0$ under this translation?

A. $3x+y-18=0$ C. $3x-18=0$

B. $3x-2y-18=0$ D. $3x+2y-18=0$

24. The image of point $P(4,0)$ when it has been rotated through an angle $\theta=135^{o}$ about the origin is given as:

A. $(-2\sqrt{2},\ -2\sqrt{2})$ C. $(-2\sqrt{2},\ 2\sqrt{2})$

B. $(2\sqrt{2},\ 2\sqrt{2})$ D. $(\sqrt{2},\ -2\sqrt{2})$

25. Which variable is discrete?

A. Size of a shirt C. Price of a kilo of sugar

B. Height of students in a class D. Number of rooms in a house

26. The following table is the frequency distribution of a certain set of data.

X	10	11	12	13	14	15
F	2	0	Y	2	4	2

If the mean of the data is 12.8, then the frequency of value 12, that is y is equal to:

A. 5 B. 6 C. 4 D. 3

27. What is the term independent of x in the expansion of $\left(x^2+\frac{1}{x}\right)^6$?

A. 4 B. 15 C. 8 D. 10

28. If $\sum_{k=1}^{n} a_k = 3-2k$, then the sum of the first 10 terms is equal to:

A. 80 B. 84 C. −80 D. 24

29. Three numbers are in geometric sequence and their sum is 26 and their product is 216. Then the numbers are:

A. 8, 9, 9 C. 2, 6, 18

B. 8, 6, 12 D. 10, 10, 6

30. What is the value of k if $4^k.4^{k^2}.4^{k^3}\ldots = 16$?

A. $\frac{2}{3}$ B. $\frac{3}{2}$ C. $\frac{1}{3}$ D. $\frac{4}{3}$

31. Given the sequence $\{3, 6, 12, 24, \ldots\}$. The 10th partial sum S_{10} of the sequence is:

A. 3075 B. 1027 C. 1023 D. 3069

32. Which of the following is <u>true</u> about the given infinite series?

A. $\sum_{n=1}^{\infty}\left(\frac{4}{3}\right)^n$ is convergent C. $\sum_{n=1}^{\infty}(-1)^n\left(\frac{2}{3}\right)$ is divergent

B. $\sum_{n=1}^{\infty} 2^n$ is divergent D. $\sum_{n=1}^{\infty}(-1)^n 2^n$ is convergent

33. If n arithmetic means are inserted between 4 and 20 and the sum of the sequence after inserting the means is 120, then what is the value of n?

A. 8 B. 6 C. 9 D. 10

34. The fractional form of $0.\overline{47}$ is:

A. $\frac{47}{99}$ B. $\frac{47}{90}$ C. $\frac{47}{990}$ D. $\frac{47}{9}$

35. Suppose a ball is dropped from a height of 60 meters and always rebounces $\frac{2}{3}$ of the height from which it falls. What is the vertical distance that the ball traveled, in meters (assume that the ball will never stop bouncing)?

A. 150 B. 250 C. 300 D. 420

36. The maximum value of the function $f(x) = x^3 - 12x$ on the interval $[-4, 3]$ is equal to:

A. −2 B. −16 C. 16 D. −9

37. The area of a region R bounded by the graph of $f(x) = -x^3$, the x-axis, the lines $x = -2$ and $x = 2$ (in sq. unit) is:

A. 16 B. 8 C. 4 D. 10

38. The equation of the tangent line to the circle $(x+1)^2 + (y-2)^2 = 4$ at the point $(-1, 0)$ is given as: A. $y = 0$ B. $x = 0$ C. $y = -x$ D. $y = 2x$

39. Let $f(x) = (1-2x)^{\frac{1}{2}}$. What is the value of $f'(-4)$?

A. $\frac{-1}{3}$ B. $\frac{-1}{2}$ C. -1 D. -3

40. Which of the following are critical numbers of the function $f(x) = 4x^3 - 5x^2 - 8x + 20$?

A. $\frac{1}{2}$ and $\frac{3}{4}$ C. $\frac{-1}{2}$ and $\frac{3}{4}$

B. $\frac{-1}{2}$ and $\frac{-4}{3}$ D. $\frac{-1}{2}$ and $\frac{4}{3}$

41. The value of the definite integral given by $\int_{-2}^{3}(x+2)dx$ is:

A. $\frac{25}{2}$ B. 12 C. $\frac{21}{2}$ D. 18

42. Let f and g are two differentiable functions and $f'(-2) = -6$, $g'(-2) = 1$. Then the value of $h'(-2)$, if $h(x) = 3f(x) - 2g(x) - 1$ is:

A. -20 B. -21 C. 36 D. 101

43. If f and g are given integrable functions on the interval $[a, b]$, $k \in \mathbb{R}$ and $c \in [a, b]$, then which of the following is <u>not true</u>?

A. $\int_a^a f(x)dx = 0$ C. $\int_a^b kf(x)dx = k\int_a^b f(x)dx$

B. $\int_a^b f(x)dx = \int_b^a f(x)dx$ D. $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

44. The standard deviation is a measure of:

A. Central tendency C. Position

B. Spread of data D. Probability

45. The mean of the numbers 4, 6, 8, 10, and x is 8. Then what will be the value of x?

A. 10 B. 16 C. 14 D. 12

46. The following table shows a frequency distribution of score on final examination in Mathematics.

Scores	30__39	40__49	50__59	60__69	70__79	80__89	90__100
№ of students	1	3	11	21	43	32	9

What is the value of Q_3, (third quartile)?

A. 84 B. 70 C. 82.9 D. 74.5

47. Based on question number 46, how many students score above 75%?

A. 17.1 B. 18 C. 16.8 D. 21.1

48. A class has 40 students. The average weight is 55 kg. If a student weight 65 kg leaves, then the new mean is: A. 74.54 C. 54.74

B. 65 D. 55

49. The standard deviation of 2, 4, 6, 8 is:

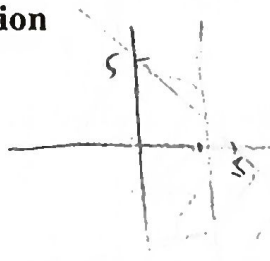
A. $\sqrt{20}$ B. 5 C. 20 D. $\sqrt{5}$

50. If the feasible region of the graph of a system of linear inequalities is unbounded, then which of the following is correct?

A. There is no solution to the objective function
B. Optimal solution always exists
C. The optimal solution may or may not exist
D. The objective function has no optimal solution

51. The maximum value of $Z = 6x + 8y$

Subject to: $\begin{cases} x + y \leq 5 \\ x \leq 3 \\ y \leq 4 \\ x \geq 0, y \geq 0 \end{cases}$ is:



A. 40 B. 42 C. 38 D. 44

52. What we call the function that to be maximized or minimized?

A. Constraint C. Feasible region
B. Objective function D. Solution set

53. The graphical method of solving linear programming (LPP) is suitable when there are:

A. Two variables C. Any number of variables
B. Four variables D. Three variables

54. The ratio between two numbers is 3: 5. If each number is increased by 4, the ratio becomes 2: 3. Then what are the original numbers?

A. 8 and 18 B. 12 and 20 C. 14 and 16 D. 12 and 24

55. If the cost price of an item is 200 Birr and sold for Birr 250, then the profit percentage is:

A. 25% B. 20% C. 30% D. 50%

56. The simple interest on Birr 5000 at 8% per year for 3 years, in Birr is:

A. 6,200 B. 1,500 C. 1,000 D. 1,200

57. A trader buys goods for Birr 15,000 and mark up them by 20% on retail. Then the selling price, in Birr is: A. 300 B. 1,800 C. 18,750 D. 1,500

58. An investment of Birr 1,000 gives a profit of Birr 200. What is the Return on Investment (RON)? A. 10% B. 20% C. 30% D. 25%

59. A woman buys a washing machine with price exclusive of VAT (15%) is Birr 24,000. If the VAT is charged, how much will the woman pay for the machine?

A. 26,500 B. 27,600 C. 36,000 D. 3,600

60. Which one of the following is a direct tax?

A. Value Added Tax, VAT
B. Turn Over Tax, TOT
C. Income tax
D. Excise tax

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E.C

BLESSED G. MICHAEL CATHOLIC SECONDARY SCHOOL

MODEL-EXAM ONE Answer Sheet **Subject:** Mathematics.

Allotted time: 1:40'

Name Hanna Ayalew **Sec.** 12thC **No.** 24.

Direction: Blacken the circles given next to the numbers as per your answer's choice.

	A	B	C	D
1.	●	O	Ø	O
2.	O	●	O	O
3.	O	O	O	●
4.	O	●	O	O
5.	●	O	O	O
6.	O	O	O	●
7.	O	O	●	O
8.	●	O	O	O
9.	O	O	●	O
10.	●	O	O	O
11.	O	O	●	O
12.	●	O	O	O
13.	O	O	O	●
14.	O	O	O	●
15.	O	●	O	O
16.	O	●	O	O
17.	O	O	●	O
18.	O	O	O	●
19.	O	O	●	O
20.	O	O	O	●
21.	●	O	O	O
22.	O	O	O	●
23.	O	O	O	●
24.	O	O	●	O
25.	O	O	O	●
26.	●	O	O	O
27.	O	●	O	O
28.	O	O	●	O
29.	O	O	●	O
30.	●	O	O	O

	A	B	C	D
31.	O	O	O	●
32.	O	●	O	O
33.	●	O	O	O
34.	●	O	O	O
35.	O	O	●	O
36.	O	O	●	O
37.	O	●	O	O
38.	●	O	O	O
39.	●	O	O	O
40.	O	O	O	●
41.	●	O	O	O
42.	●	O	O	O
43.	O	●	O	O
44.	O	●	O	O
45.	O	O	O	●
46.	O	O	●	O
47.	✓	O	O	O
48.	O	O	●	O
49.	O	O	O	●
50.	O	O	●	O
51.	O	O	●	O
52.	O	●	O	O
53.	●	O	O	O
54.	O	✓	●	O
55.	●	O	O	O
56.	O	O	O	●
57.	O	●	O	O
58.	O	●	O	O
59.	O	●	O	O
60.	O	O	●	O

	A	B	C	D
61.	O	O	O	O
62.	O	O	O	O
63.	O	O	O	O
64.	O	O	O	O
65.	O	O	O	O
66.	O	O	O	O
67.	O	O	O	O
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90.	O	O	O	O

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102.	O	O	O	O
103.	O	O	O	O
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108.	O	O	O	O
109.	O	O	O	O
110.	O	O	O	O

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Blessed G/Michael Catholic Secondary School

Mathematics Model Examination for grade 12th.

May. 2018 **Time allotted:- 3:00'**

Name ---Hanna Ayalew--- Sec. ------ No. -------

I. Choose the best answer from the given alternatives.

1. Which one of the following true about the graph of f(x) = $\sqrt{x-4}$

A. domain of f is lR

B. Range of f is lR

C. domain of f is [4,∞)

D. symmetric with respect to y-axis

2. If f(x) = $3x+\sqrt{4-x}$, then the value of f(-5) is _____

A. -7 B. 4 C. 3 D. -12

3. let f(x) = 3-x and g(x) = -x-4, then which one of the following is the value of (2f+3g)(1) is _____

A. 11 B. -11 C.13 D. 2

4. Let f(x) = $-x^2+6x+7$, then which one of the following is the vertex of the graph of f:

A. (3,16) B. (4,13) C. (1,7) D. (0,7)

5. If for sets A,B and C, AUC = {1,2,3,6} BUC= {1,8,7,6,3} then (AnB)uC) is equal to

A. {1,8,7,6,3} B. {1,3,6} C. {1,2,3,6} D. {8,7}

6. If one of the roots of quadratic x2-8x+k = 0 exceeds the other by 6, then the value of x is _____

A. 7 B. 1 C. 3 D. 4

7. $\frac{1}{4\sqrt{2}}\left(\frac{-\sqrt{2}+1}{\sqrt{2}-1}+\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)$ *is* _____

A. 0 B. -1 C. 1 D. 4

8. For what number A is x-1 a factor of f(x) = $2x^5-Ax^3+5x-1$

A. 4 B. - 4 C. 6 D. -2

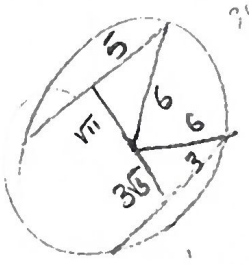
9. Two buildings A and B on a level ground are 6 meters apart as seen from the top of building A that is 10 meters high the angle of elevation of the top building B is 30^0 how high is building B in meters:

A. $2\sqrt{3}$ *B.* $4+2\sqrt{3}$ *C.* $2\sqrt{3}+10$ *D.* $2\sqrt{3}-4$

10. If cos = $\frac{4}{5}$ and $0<\theta<\frac{\pi}{2}$, then $\csc^2\theta$ is equal to

A. $\frac{3}{5}$ *B.* $\frac{9}{25}$ *C.* $\frac{25}{9}$ *D.* $\frac{4}{3}$

11. In a circle of radius 6cm, there are two parallel chords of length 10cm and 6cm the distance between the chord is



A. $2\sqrt{5}+11$ *B.* $4+\sqrt{11}$ *C.* $4+\sqrt{5}$ *D.* $-4+\sqrt{5}$

12. If n(A) = 10, n(B) = 7 and n(C)= 4 for three disjoint sets A,B and C, then (AUBUC) = _____

A. 280 B. 13 C. 17 D. 21

13. If x and y are two positive integers both dividable by 5, then which one of the following is not necessary true?

A. x+y is dividable by 5

B. x-y is divisible by 5

C. x^2-y^2 is dividable by 5

D. x+y is divisible by 5

14. Which one of the following is the number of diagonals in an octagon?

A. 8 B. 20 C. 14 D. 30

15. If the radius of a regular hexagon is 8cm, then which one of the following is the apothem of the polygon.

A. $4\sqrt{3}$ B. 8 C. $3\sqrt{2}$ D. $4\sqrt{2}$

16. The area of regular 12 sided polygon is $48cm^2$ then what is the area of circle.

A. $12\pi cm^2$ B. $36\pi cm^2$ C. $16\pi cm^2$ D. $32\pi cm^2$

17. The Solution set of $\log [2+log_3^{(x+3}] = 0$ is equal to

A. $\frac{10}{3}$ B. $\frac{-4}{3}$ C. $\frac{-8}{3}$ D. $\frac{1}{3}$.

18. If $8^{3x-1} = (128)^{-x-1}$, then what is the value of x ?

A. $\frac{-1}{4}$ B. $\frac{-4}{15}$ C. $\frac{-1}{2}$ D. $\frac{1}{2}$

19. Which one of the following is true about $f(x)=\frac{4x^3-4x^2+4x}{3}$

A. it is polynomial function

B. The constant term is 3

C. The leading coefficient is $\frac{4}{3}$

D. The degree of x^2 is - 4

20. If the angle of the sector with radius 6cm is 240^0 then the area of the sector is _____

A. $48\pi cm^2$ B. $24\pi cm^2$ C. $12\pi cm^2$ D. $10\pi cm^2$

21. Which one of the following is the equation of the line which is parallel to y= 3x+4 and passing through the point (0,2) is _____

A. y = 3x+4 B. y = 3x+2 C. y = 3x-2 D. y = 3x-4

22. Which one of the following functions are inverse each other

A. $f(x)=\frac{x+2}{x-2}, g(x)=\frac{2x+1}{x+1}$

B. $f(x)=\sqrt{x}\, g(x)=|x|$

C. f(x)=8x and $g(x)=\frac{x}{8}$

D. $f(x)=log_2^x, g(x)=\ell nx$

23. Which one of the following is true about the function $f(x)=\frac{x+1}{x^2+5x+6}$

A. Its vertical Asymptote is x=2

B. y= 0 is horizontal asymptote

C. has oblique asymptote

D. a hole point is at x= -2

24. $\begin{vmatrix} -1 & 3 & -6 \\ 1 & -2 & 1 \\ x & 3 & 0 \end{vmatrix} = 1$, then the value of x is equal to

A. $\frac{16}{9}$ B. $\frac{-16}{9}$ C. 24 D. 3

25. Let A= (aij) 4x3 and B= (bij) mxn, which one of the following is the value of M and n for AB defined

A. m=3, n=4 only

B. m=3,n is any positive integers

C. m= 3, n= 2

D. m= 3, n= 4

26. In a triangle the smaller angle measure 20^0 less than half of the largest angles, the middle angle measures 30^0 more than the smallest angle the measure of each angle of the triangle is _____

A. 27.5,57.5,95 B. 35,45,90 C. 30,60,90 D. 40,90,70

The determinant of $\begin{bmatrix} d & e & f \\ a & b & c \\ g & h & k \end{bmatrix}$ is _____

A. 4 B. 16 C. -4 D. 8

28. The inverse of matrix A= $\begin{bmatrix} 1 & 3 & 0 \\ -1 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$ is _____

A. $\begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{1}{4} & 2 \end{bmatrix}$ B. $\begin{bmatrix} \frac{1}{4} & \frac{-3}{4} & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 \\ \frac{-1}{8} & \frac{-1}{8} & \frac{1}{2} \end{bmatrix}$, C. $\begin{bmatrix} 0 & 1 & 3 \\ 1 & -1 & 2 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} \end{bmatrix}$ D. $\begin{bmatrix} \frac{1}{8} & \frac{1}{4} & 0 \\ 3 & 1 & 2 \\ \frac{3}{4} & \frac{1}{8} & 2 \end{bmatrix}$

29. If the area of a triangle with vertical (-2,1) (2,0) and (k,2) is 16 square units the value of k is _____

A. k = 26 B. k = 38 C. k = 4 D. k = 12

30. Let U= (2,1) and v= (-4,6) then the value of 2U+4V is _____

A. (4,16) B. (-12,26) C. (14,12) D. (4,-12)

31. If $|U| = 4$ *and* $|V| = 2$. Vector U and V makes an angle $\theta = \frac{3\pi}{2}$ between them then the value of $|2U + V|$ *is* _____

A. $\sqrt{17}$ B. $2\sqrt{17}$ C. $3\sqrt{14}$ D. $3\sqrt{17}$

32. If U= i+2j+2k and V= 3i+j-2k,then find the cross product of VxU is _____

A. 6i-8j+k B. 6i-8j+3k C. 6i-8j+5k D. 3i-2j+k

33. Which one of the following is the area of a parallelogram for the vector U= 2i+4j+k and V= -4i-3j+2k is _____

A. $\sqrt{200}$ B. $\sqrt{128}$ C. $\sqrt{145}$ D. $\sqrt{285}$

34. Cosider the line is passing through the point (2,3) and (4,6) then the parametric equation of the line is ____

A. x=2+2t B. x= 2-2t C. x=2-2t D. x=2+3t

Y=3+3t y= 3-2t y= 1-4t y= 1-3t

35. The equation of the tangent line to the circle of $x^2+y^2+10x+2y = -13$ a point (-3,2) is _____

A. y = -2x+4 B. $y = \frac{-2}{3}x$ C. $y = 4x + 4$ D. $y = 2x - 4$

36. A ballon is rising vertically at 5m/se, wind is blowing from east to west at 4m/se. the resultant velocity of the bollon is _____

A. $\sqrt{45}$ B. $\sqrt{40}$ C. $\sqrt{41}$ D. $\sqrt{24}$

37. A translation T takes the circle $x^2+y^2+2x+4y=0$ in to the circle x2+y2-4x+2y=0 then what is the translatio vector of the image

A. (3,2) B. (3,1) C. (1,2) D. (-2,1)

38. Which one of the following is the image of the line ℓ. : 3x+2y=6 after rotation of $\frac{\pi}{2}$ *about the point* (2,3 is _____

A. 2x-3y = 6 B. 3x-2y =-6 C. 3x-2y=1 D. 2x-3y=1

39. If a translation T takes the circle $x^2+y^2-2x+6y+3=0$ in to the circle whose equation $(x+2)^2+(y-4)$ then what is the image of the origin under T.

A. (-3,7) B. (1,2) C. (1,-3) D. (-2,4)

40. The image of the line y= 2x+5 after it has been reflacted along the line $y = \frac{-1}{2}x-4$ is _____

A. y = x+5 B. y = 2x+5 C. y = 2x-4 D. $y = \frac{-1}{2}x+4$

41. Which one of the following is true about the given data

Score	Frequency
$10 \leq t < 15$	3
$15 \leq t < 20$	1
$20 \leq t < 25$	4
$25 \leq t < 30$	2

A. D5= 21.25

B. Q3= 21.17

C. IQR = 8.3

D. MD = 24

42. A box of 10 candles of 4 defective and 6 non- defective candles. If 5 of these candle are selected at randon What is the probability in which 3 will be defective?

A. $\frac{5}{14}$ B. $\frac{5}{24}$ C. $\frac{5}{21}$ D. $\frac{4}{22}$

43. The radius of aright circular cylinder of height 20 cm increase at rate of 4cm/min. what is the rate of change of the volume of cylinder when the radius is 6m?

A. $960\pi cm^2/\min$ B. $140\pi cm^2/\min$ C. $120\pi cm^2/\min$ D. $400\pi cm^2/min$

44. Which one of the following is the equation of the tangent line at appoint where the function $f(x) = x^3-x^2-x+1$ cross : The x- axis:

A. y = 2x+2 B. y = 4x+4 C. y = 3x-4 D. y = 3x+4

45. Which one of the following is not true

A. $\int_1^3 2x + 3\,dx = 14$

B. $\int_a^b f(x)dx - \int_c^b f(x)dx = \int_a^c f(x)dx$

C. The area of the graph of $f(x) = x^3-9x$ for [-2,2] is 14

D. $\frac{d}{dx}\int f(x)dx = f(x)$

46. Which one of the following is not true about the following data

Score	0-10	10-20	20-30	30-40	40-50
Frequency	3	1	1	2	3

A. Mean = 26 B. median = 30 C. P_{25}= 10.3 D. D_5= 30

47. Which one of the following sequence is an arithmetic sequence

A. $\alpha n = 4$ B. $\alpha n = 2^n$ C. $\alpha n = \left(\frac{1}{2}\right)^n$ D. $\alpha n = cos(n\pi)$

48. If the nth term of an arithmetic sequence is x and its $(3n)^{th}$ term is y then find the value of $(2n)^{th}$ tem of the sequence .

A. x-y B. x+y C. $\frac{x+y}{2}$ D. $\frac{x-y}{2}$

every 6 months, during his/her 6 year on the job. If his/her starting salary is Birr 2000 per month what will his/her annual salary be at the end of 4th year.

A. 3400 B. 31,800 C. 15,800 D. 16,200

50. The sum of two digit number which leave the remainder 3 when divided by 5 is _____

A. 45 B. 99 C. 495. D. 999.

51. Which one of the following is the value k for $\sum_{n=1}^{k}(2n + 5) = 520$

A. 20 B. 10 C. 3 D. 15

52. A ball is dropped from a height of 40meter In each bounce. If redounds $\frac{1}{3}$ of the height from which it falls the total distance from the ball travels up and down is _____

A. 80 B. 40 C. 60 D. 120

53. Which one of the following is convergent sequence

A. $(-1)^n$ B. $\left(\frac{1}{2}\right)^n$ C. 3^n D. $\left(\frac{4}{3}\right)^n$

54. The rate change of the function $f(x)= x^2-3$ from [-2,2] is

A. $\frac{1}{2}$ B. 4 C. 0 D. ∞

55. The biomass of a bacterial culture in milligram as a function of time in minutes is given of $f(t) = 4t^3+9$, what is the instantaneous rate of growth of the culture when t= 2 minutes?

A. 41 B. 46 C. 48 D. 96

56. Which one of the following is true about f and g are differentiable at x .

A. $(fg)'(x) = f'(x) + g'(x)$ C. $(fg)'(x)=f'g(x) + g'(x)f(x)$

B. $\left(\frac{f}{g}\right)'(x) = \frac{f'(x)-g'(x)}{(f(x)^2}$ D. $\left(\frac{f}{g}\right)'(x) = f'(x)g(x) - g'(x)f(x)$

57. Which one of the following is the value of k so that the line $y = -2x+k$ is tangent to the curve $y= 2x^2+2x+1$

A. 2 B. 1 C. 4 D. -1

58. If $h(x) = x\, f(x^2+4x)$, $f(12) = -4$ and $f'(12) =4$ then $h'(2)$ is equal to: -

A. 40 B. 32 C. 28 D. 44

59. Which one of the following is true about the graph of $f(x)= x^3-3x^2+3x+1$

A. f is strictly decreasing on $[-\infty, \infty)$ C. *f has local extrem value at* $x = 1$

B. f is strictly increasing on $(-\infty, \infty)$ D. *absulte maximum value is* 40

60. Which one of the following is the maximum value of $Z= 2x+4y$

$$\text{Subject} \begin{cases} 2y + x \geq 10 \\ y \geq -3x + 10 \\ x \geq 0, y \geq 0 \end{cases}$$

A. 40 B. 20 C. 30 D. 60

61. The term of loan if the principal is Birr 20000 the interest rate is 8% and total interest is Birr 320 is _____

A. 72 day B. 48 day C. 96 day D. 12 day

$f(12) + (f'(12)(2x+4))x$
$-4 + (4+8)2$
$-4 + 64 = 60$

55/60

BLESSED G. MICHAEL CATHOLIC SECONDARY SCHOOL

MODEL-EXAM Answer Sheet **Subject:** Mathematics.

Name Hanna Ayalew **Allotted time: 2:10hr**

Sec. 12TC **No.** 21.

Direction: Blacken the circles given next to the numbers as per your answer's choice.

	A	B	C	D
1.	O	O	●	O
2.	O	O	O	●
3.	O	●	O	O
4.	●	O	O	O
5.	O	●	O	O
6.	●	O	O	O
7.	O	O	●	O
8.	O	O	●	O
9.	O	O	●	O
10.	O	O	●	O
11.	O	●	O	O
12.	O	O	O	●
13.	O	O	●	O
14.	●	O	O	O
15.	●	O	O	O
16.	O	O	●	O
17.	O	O	●	O
18.	●	O	O	O
19.	O	O	●	O
20.	O	●	O	O
21.	O	●	O	O
22.	O	O	●	O
23.	O	●	O	O
24.	O	●	O	O
25.	O	●	O	O
26.	●	O	O	O
27.	O	O	●	O
28.	O	●	O	O
29.	●	O	O	O
30.	O	●	O	O

	A	B	C	D
31.	O	●	O	O
32.	O	O	●	O
33.	O	O	O	●
34.	●	O	O	O
35.	O	●	O	O
36.	O	O	●	O
37.	O	●	O	O
38.	O	O	O	●
39.	●	O	O	O
40.	O	●	O	O
41.	●	O	O	O
42.	O	O	●	O
43.	●	O	O	O
44.	O	●	O	O
45.	O	O	●	O
46.	O	O	●	O
47.	●	O	O	O
48.	O	O	●	O
49.	O	●	O	O
50.	O	●	O	●
51.	●	O	O	O
52.	●	O	O	O
53.	O	●	O	O
54.	O	O	●	O
55.	O	O	●	O
56.	O	O	●	O
57.	O	O	O	●
58.	O	O	●	O
59.	O	O	●	O
60.	●	O	O	O

	A	B	C	D
61.	●	O	O	O
62.	O	O	O	O
63.	O	O	O	O
64.	O	O	O	O
65.	O	O	O	O
66.	O	O	O	O
67.	O	O	O	O
68.	O	O	O	O
69.	O	O	O	O
70.	O	O	O	O
71.	O	O	O	O
72.	O	O	O	O
73.	O	O	O	O
74.	O	O	O	O
75.	O	O	O	O
76.	O	O	O	O
77.	O	O	O	O
78.	O	O	O	O
79.	O	O	O	O
80.	O	O	O	O
81.	O	O	O	O
82.	O	O	O	O
83.	O	O	O	O
84.	O	O	O	O
85.	O	O	O	O
86.	O	O	O	O
87.	O	O	O	O
88.	O	O	O	O
89.	O	O	O	O

	A	B	C	D
91.	O	O	O	O
92.	O	O	O	O
93.	O	O	O	O
94.	O	O	O	O
95.	O	O	O	O
96.	O	O	O	O
97.	O	O	O	O
98.	O	O	O	O
99.	O	O	O	O
100.	O	O	O	O
101.	O	O	O	O
102.	O	O	O	O
103.	O	O	O	O
104.	O	O	O	O
105.	O	O	O	O
106.	O	O	O	O
107.	O	O	O	O
108.	O	O	O	O
109.	O	O	O	O
110.	O	O	O	O
111.	O	O	O	O
112.	O	O	O	O
113.	O	O	O	O
114.	O	O	O	O
115.	O	O	O	O